y is known, A is known,
what is X and f?
F can be chosen from a list of predefined logarithm

X

Student	kindness	Handsomeness	Beautiful smile	How old?
An Duc	10	8	1	20
Kien	10	7	4	16
Hieu	10	1	10	20

W

W
w1
w2
w3
w4

Y

Consider as boy friend
1
1
0

Sigmoid (x) = 1/(1 + e^-x)
sigmoid (f(x1, x2)) =
1 / (1 + e^(-f(x1, x2)))
= 1 / (1 + e^(-(2x1 + x2))

f(K1 * **x1** + h1 * **x2** + b1 * **x3**) = 1

f(K2 * **x1** + h2 * **x2** + b2 * **x3**) = 1

f(K3 * **x1** + h3 * **x2** + b3 * **x3**) = 1

f(x, A1) = 1

f(x, A2) = 1

f(x, A3) = 0

F(x1, x2) = 2x1 + x2

F (AX) = A.X

X

Index	Student	kindness	Handsomeness	Beautiful smile	How old?
0	An Duc	10	8	1	20
1	Kien	10	7	4	16
2	Hieu	10	1	10	20

LEARN?

W
w1
w2
w3
w4

Y

Consider as boy friend
1
1
0

$$\begin{cases} 1/[1 + e^{-(10w1 + 8w2 + 1w3 + 20w4 + w5)}] = 1 \\ 1/[1 + e^{-(10w1 + 7w2 + 4w3 + 16w4)}] = 1 \\ 1/[1 + e^{-(10w1 + 1w2 + 10w3 + 20w4)}] = 0 \end{cases}$$

Sigmoid (x) = 1/(1 + e^-x)

F(x1, x2) = 2x1 + x2

—> y = sigmoid (f(x1, x2)) = 1 / (1 + e^(-f(x1, x2))) = 1 / (1 + e^(-(2x1 + x2)) —> what is the value range of y? —> Answer: (0, 1).

AnDuc: Input : [10, 8, 1, 20] , output: y = 1

Sigmoid (10w1 + 8w2 + 1w3 + 20w4) = 1, what are the values of w1, w2, w3, w4 that satisfies this?

Sigmoid (10w1 + 8w2 + 1w3 + 20w4) = 1 / (1 + e^-(10w1 + 8w2 + 1w3 + 20w4))

In another word, f(W, X[0,:]) = 10w1 + 8w2 + 1w3 + 20w4 and we need to find W vector such that.

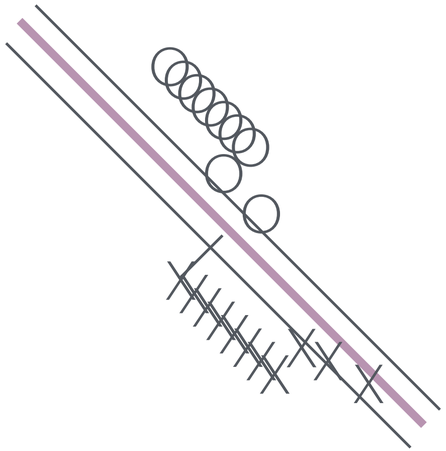
y[0] = sigmoid(f(W, X[0, :])) = 1

y[1] = sigmoid(f(W, X[1, :])) = 1.

y[2] = sigmoid(f(W, X[2, :])) = 0

$$\begin{cases} 1/[1 + e^{-(10w1 + 8w2+1w3 + 20w4)}] = 1 \\ 1/[1 + e^{-(10w1 + 7w2 + 4w3 + 16w4)}] = 1 \\ 1/[1 + e^{-(10w1 + 1w2 + 10w3 + 20w4)}] = 0 \end{cases}$$

F (AX) = A.X



House Number	X_0	X_1 Total Area (squared	X_2 How old (Year)	Y House Price
1	1	1000	10	700,000
2	1	2000	19	1,000,000
3	1	3000	29	1,200,000
4	1	3400	46	800,000

1	1000	10
1	2000	19
1	3000	29
1	3400	46

700,000
1,000,000
1,200,000
800,000

h_theta(x) = theta_0 + theta_1 X_1+ theta_2 X_2
= theta_0 * X_0 + theta_1 X_1+ theta_2 X_2
= X Theta
— > Theta^T X —> 1, 3 , X^T —> y = y^T
where
theta =
[theta_0,
theta_1,
theta_2]
where
X =
[X_00, X_01, X_02
X_10, X_11, X_12,
X_20, X_21, X_22,
X_30, X_31, X_32,] —> 4 rows, 3 columns
—> X's size is (4,3)
Y = [700k,
1,000k
1,200k
800k] —> 4 rows , 1 column —> (4, 1) shape

House Number	X_0	X_1 Total Area (squared	X_2 How old (Year)	Y House Price
0	1	1000	10	700,000
1	1	2000	19	1,000,000
2	1	3000	29	1,200,000
3	1	3400	46	800,000

h_theta(x) = Theta X —>
cost_function (objective function):
= MSE (Mean Squared Error) = 1/N (Theta X ^2 - Y)*2
= 1/2N [
(X[0, :] Theta - Y[0])^2 +
(X[1, :] Theta - Y[1])^2 +
(X[2,:] Theta - Y[2])^2
]

$$= \sum_{i=1}^{i=N} \frac{1}{2N}(h_\theta(x_i) - y_i)^2 \quad = \sum_{i=1}^{i=N} \frac{1}{2N}(ThetaX_i - Y_i)^2$$

Steps:
- measure the gradient of the cost with regard to the parameter.
- update the parameter
- Repeat until the lost/cost is minimized at a satisfied point.

House Number	X_0	X_1 Total Area (squared	X_2 How old (Year)	Y House Price
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3	1	3400	46	800,000

Steps:

- measure the gradient of the cost with regard to the parameter.
- update the parameter
- Repeat until the lost/cost is minimized at a satisfied point.

$$J = \sum_{i=1}^{i=N} \frac{1}{2N}(h_\theta(x_i) - y_i)^2 \quad = \sum_{i=1}^{i=N} \frac{1}{2N}(ThetaX_i - Y_i)^2$$

$$\frac{dJ}{d\theta} = \frac{1}{2N} \sum_{i=1}^{i=N} \frac{d(\theta X_i - y_i)^2}{d\theta} \quad \frac{dJ}{d\theta} = \frac{1}{2N} \sum_{i=1}^{i=N} \frac{d(\theta X_i - y_i)^2}{d\theta}$$

$$J = \sum_{i=1}^{i=N} \frac{1}{2N}(\theta X_i - Y_i)^2$$

$$\frac{dJ}{d\theta}$$

Gradient descent method:

$$\theta_{new} = \theta_{current} - \alpha * \frac{dJ}{d\theta}$$

α : learning rate

J = x^2 + 3

J = 1/2(x^2 + y^2) —> find x and y to minimize J.
dJ/dx = x
dJ/dy = y
Step 0: Initially X = 9, Y = 8, alpha = 0.1
—> J = 1/2(9**2 + 8**2) =145 /2 = **72.5**
Step 1:
X-new = X-current - 0.1 * X_current
Y-new = Y-current - 0.1 * Y_current

X_new = 9 - 0.1 * 9 = 8.1
Y_new = 8 - 0.1*8 = 7.2
—> J = 1/2 (8.1**2 + 7.2**2) = **58.725**

Step 2:
X-new = X-current - 0.1 * X_current
Y-new = Y-current - 0.1 * Y_current

X_new = 8.1 - 0.1*8.1 =7.29
Y_new = 7.2 - 0.1*7.2 = 6.48

α : learning rate

Step decay:

$$\alpha_{current} = \alpha_{original} * \text{drop_rate}^{\frac{epoch}{\text{step_size}}}$$

Exponential decay:

$$\alpha_{current} = \alpha_{original} * \exp{-\text{decay_rate} * \text{epoch}}$$

Inverse time decay

$$\alpha_{current} = \alpha_{original} * \frac{1}{(1 + decay * epoch)}$$

Polynomial decay

$$\alpha_{current} = \alpha_{original} * \frac{1}{(1 - \frac{epoch}{maxEpoch})}^{power}$$

Mean Squared Error (MSE) Loss function:

$$J = \sum_{i=1}^{i=N} \frac{1}{2N}(\theta X_i - Y_i)^2$$

$$\frac{dJ}{d\theta} = \frac{1}{2N} \sum_{i=1}^{i=N} 2(\theta X_i - Y_i) \frac{d(\theta X_i - Y_i)}{d\theta}$$

$$\frac{dJ}{d\theta} = \frac{1}{N} \sum (\theta X_i - Y_i) X_i$$

Homework:
J = 1/2 (X^2 + Y^2)

Theta = [X, Y]
J = 1/2 Sum(theta**2) = 1/2 theta theta^T

dj/dtheta = theta

Update theta:
- theta_new = theta - alpha * dj/dtheta

= theta - alpha*theta

Mean Squared Error (MSE) Loss function:

$$J = \sum_{i=1}^{i=N} \frac{1}{2N}(\theta X_i - Y_i)^2$$

L1 regularization

$$J = \sum_{i=1}^{i=N} \frac{1}{2N}(\theta X_i - Y_i)^2 + \beta_1 * \|w\|$$

$$\|w\| = \sum |w_i|$$

L2 Regularization (Ridge Regularization)

$$J = \sum_{i=1}^{i=N} \frac{1}{2N}(\theta X_i - Y_i)^2 + \frac{\beta_2}{2} * \|w\|^2$$

$$\|w\| = \sum w_i^2$$

L1 + L2 regularization (Elastic Net)

$$J = \sum_{i=1}^{i=N} \frac{1}{2N}(\theta X_i - Y_i)^2 + \beta_1 * \|w\| + \frac{\beta_2}{2} * \|w\|^2$$